DATABASE MANAGEMENT SYSTEMS
MODULE 5

Microsoft Access: forms, data entry, input masks and validation

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Direct Answer. Exam strategy. The supplied Module 5 page lists learning materials rather than lecture slides. The teacher expects direct questions, so this note set is organized as definitions, contrasts, steps, examples and short model answers on the listed topics. Start by memorizing the bold definitions and the comparison table.

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Microsoft Access Basics

What is Microsoft Access?

Microsoft Access is a relational database management system (RDBMS) in Microsoft Office. It lets users create and manage a database using tables, queries, forms, reports, macros and (optionally) VBA modules. An Access database stores related data and the objects used to enter, retrieve and present it.

- **Table:** stores data in rows (records) and columns (fields).
- **Query:** retrieves, filters, sorts, calculates or changes data according to criteria.
- **Form:** provides a convenient screen for entering, viewing and editing data.
- **Report:** formats data for viewing, printing or sharing.
- **Macro/module:** automates actions; modules contain VBA code.

What are fields, records and data types?

A **field** is one attribute or column, such as StudentName. A **record** is one complete row containing values for the fields, such as one student's details. A **data type** determines what kind of values a field can store and constrains input. Examples include Short Text, Number, Date/Time, Currency, Yes/No and AutoNumber. A primary key uniquely identifies each record.

> **Direct Answer. Q: What is the difference between a field and a record?**
> **A:** A field is a column that stores one type of attribute for every record; a record is a row containing the complete set of field values for one item or person.

Forms

What is a form? Why is it used?

A form is a database object that provides a user-friendly interface to the data in tables or queries. A bound form is connected to a record source, so users can view, enter, edit and delete records according to its settings. Forms can arrange controls clearly, show instructions, limit the available operations and apply input rules. Forms are generally easier and safer for routine data entry than editing a table directly.

Common form controls

- **Text box:** displays or accepts a value, commonly bound to a field.
- **Label:** displays descriptive text.
- **Combo box:** lets the user choose from a list or enter a value, depending on settings.
- **Check box:** represents a Yes/No value.
- **Command button:** runs an action such as opening a form or saving a record.
- **Subform:** displays related records inside a main form, often for one-to-many data.

How can a form be created?

A common method is the Form Wizard:

1. Select a table or query as the form's record source.
2. Choose the fields to display.
3. Choose a layout, such as Columnar, Tabular or Datasheet.
4. Select a style if offered.
5. Enter a form title and finish.
6. Open the form in Form view to use it, or Layout/Design view to modify its appearance and controls.

Forms can also be created with the Form command, Blank Form, Multiple Items, Split Form or Form Design. A main form with a subform is useful when one parent record is related to several child records.

Exam-ready Answer. Q: Explain how to create a form using the Form Wizard. Open the Create tab and choose Form Wizard. Select the source table or query, move the required fields into the selected list, choose a layout and style, enter a title, and finish. Open the form in Form view to enter or browse records. Use Layout or Design view to adjust controls and appearance.

Form views

View	Purpose
Form View	Enter, view and edit records.
Layout View	Change the form's layout while seeing live data.
Design View	Change detailed structure and properties; data is not shown as the design surface.

Data Entry and Modification

How do you add, edit and delete a record?

In a table Datasheet view or a bound form, move to the new-record row/button, enter field values, then move to another record or save the record. To edit, locate the record and change the required value. To delete, select the record and use Delete Record, then confirm. Deletion can be restricted when related records or referential-integrity rules prevent it.

- Use a form for guided, routine entry and clearer instructions.
- Use a table Datasheet view for direct inspection or small administrative changes.
- Use a query for targeted retrieval or controlled bulk updates/deletes, checking its criteria carefully first.

What does data entry and modification mean?

Data entry means adding new records to a database. Data modification means changing existing values or records. Deleting a record removes it from the table. Access can perform these operations directly in tables or through forms and action queries.

Do Not Confuse. A record is usually saved when you move away from it or explicitly save it. Before deleting or running an update/delete query, verify that the selected record or criteria are correct.

Input Masks

Definition and purpose

An input mask is a pattern that controls the format in which data is entered. It is useful for values with a consistent format, such as telephone numbers, postal codes or dates. It guides or constrains entry; it does not change the field's data type, and it does not by itself change how the value is stored.

Three parts of an input mask

An input mask can contain up to three parts separated by semicolons:

1. **Format pattern (required):** mask characters, placeholders and literal punctuation.
2. **Storage option (optional):** `0` stores literal mask characters with the entered data; `1` displays literals but does not store them.
3. **Placeholder (optional):** one character used to show where input goes; underscore is the default.

Example: `(000) 000-0000;0;-` requires digits in all ten positions, stores the punctuation and uses a hyphen as the placeholder. A mask is not a substitute for choosing an appropriate field data type.

Common input-mask characters

Character	Meaning	Example use
0	Required digit	`00000`
9	Optional digit	Optional part of phone number
L	Required letter	Required alphabetic position
?	Optional letter	Optional alphabetic position
A	Required letter or digit	Required alphanumeric position
a	Optional letter or digit	Optional alphanumeric position
#	Optional digit, space, plus or minus sign	Signed number pattern
¿	Convert following letters to uppercase	`>LLL`
¡	Convert following letters to lowercase	
!	Fill mask from left to right	Phone mask
\	Display following character literally	Literal symbol

How do you set an input mask?

Open the table in Design View, select the field, and find Input Mask in Field Properties. Click the build button to open the Input Mask Wizard. Choose and test a mask, select whether literal characters should be stored, finish and save. A custom mask can be typed directly into the property. A form or report control can also have an Input Mask property.

> **Direct Answer. Direct question: Give an input mask for a telephone number.**
> A common US-style pattern is `(000) 000-0000`. Each `0` requires one digit. A mask with `9` in part of the pattern makes those digit positions optional. The exact format should match the required phone-number convention.

Validation Rules and Validation Text

What is a validation rule?

A validation rule is an expression that defines which values are acceptable. Access checks the entered value against the rule. If the expression evaluates as false, Access rejects the value. Rules can apply to a field, a whole record, or a control on a form.

Field rule, record rule and form control rule

Type	What it checks	Example
Field validation	One field's value	`>=0` for nonnegative price
Record validation	Relationship between fields in one record	`[EndDate]>=[StartDate]`
Form control validation	Value entered in a particular form control	An age rule that applies only on one form

What is validation text?

Validation Text is the user-facing message Access displays when a validation rule is broken. A good message states what is wrong and how to correct it. For example, for `>=0`, use "Enter zero or a positive amount."

Examples of validation rules

Rule	Meaning
`>=0`	Value must be zero or positive.
`Between 0 And 100`	Value must lie between 0 and 100 inclusive.
`<Date()`	Date must be earlier than today.
`[EndDate]>=[StartDate]`	End date cannot precede start date.
`Is Not Null`	A value is required.
`In ("A","B","C")`	Value must match one listed option.

How do you add a validation rule?

For a table field, open Design View, select the field and enter the expression in the Validation Rule property; enter a helpful message in Validation Text. A table record validation rule is entered in the table's properties and can compare fields. For a form control, open the form in Layout or Design view, select the control, open its property sheet, then fill Validation Rule and Validation Text.

> **Exam-ready Answer. Q: State the difference between Validation Rule and Validation Text.**
> Validation Rule is the condition that the entered data must satisfy. Validation Text is the explanatory message shown when the data violates that condition. Example: Rule `>=0`; Text "Enter a nonnegative amount."

Input Mask vs Validation Rule

Feature	Input mask	Validation rule
Main purpose	Controls entry pattern or format	Checks whether a value satisfies a logical condition
Example	Phone number has required digit positions	Price must be greater than or equal to zero
Checks	Character positions and formatting	Value or relationship between values
Storage	May optionally store literal characters	Does not define a storage format
Can compare fields?	No	Record rule can compare fields

> **Remember. Quick distinction:** Mask = "How must the value be typed?" Rule = "What values are allowed?" Validation Text = "What should the user do when the value is not allowed?" A field can use more than one method.

> **Do Not Confuse.** Input masks do not replace data types or semantic validation. A phone mask can enforce a digit pattern but cannot establish that the number is real. A rule like `>=0` checks a value range but does not format a phone number.

Direct Question Bank with Answers

Definitions / 2–3 marks

1. **Define Microsoft Access.** Microsoft Access is a relational database management system used to create and manage database tables and related objects such as queries, forms and reports.
2. **Define a form.** A form is a database object that provides an interface to enter, view or edit data from a table or query.
3. **Define an input mask.** An input mask is a pattern that controls the format of data entered into a field or control.
4. **Define a validation rule.** A validation rule is an expression that accepts only data satisfying a specified condition.
5. **Define validation text.** Validation text is the error message that explains why an entered value failed a validation rule.

6. **What is a record?** A record is a row in a table containing related field values for one entity or item.
7. **What is a field?** A field is a table column representing one attribute of the stored entities.

Five-mark direct answers

1. **List the main objects in Access.** Tables store records. Queries retrieve or manipulate data using criteria. Forms provide an interface for data entry and editing. Reports format data for presentation or printing. Macros and modules automate tasks.
2. **Explain the benefits of forms.** Forms make data entry easier by arranging fields in a clear layout. They can show one or more records, provide labels and instructions, offer list controls, validate entries and restrict user actions. They reduce the need to work directly in a table datasheet.
3. **Explain the parts of an input mask.** The first part defines the required pattern, placeholders and literal characters. The second optional part determines whether literal characters are stored (0) or only displayed (1). The third optional part specifies the placeholder character, which defaults to underscore. Parts are separated by semicolons.
4. **Explain field and record validation rules.** A field validation rule checks the value of an individual field, such as requiring a nonnegative price. A record validation rule checks a condition across fields in one record, such as requiring an end date on or after the start date. Validation Text can explain the condition to the user.
5. **Describe data entry and modification in Access.** Data can be added, edited and deleted in a table Datasheet view or through a bound form. To add, move to a new record and enter values. To modify, locate the record and change its values. To delete, select the record and confirm deletion. Forms are preferred for routine guided entry.

Likely short essays

1. Explain how to create a form with the Form Wizard.
2. Explain input masks with their three parts and common symbols.
3. Explain validation rules and validation text with examples.
4. Compare input masks and validation rules.
5. Explain how Access supports adding, editing and deleting records.

Last-Minute Revision

ACCESS OBJECTS
Tables store. Queries ask/change. Forms enter/edit. Reports present/print.

FORM
User interface bound to a table/query. Wizard: source, fields, layout, title, finish.

MASK
Entry pattern. 0 required digit; 9 optional digit; L required letter; ? optional letter; ¿ uppercase. Up to 3 semicolon-separated parts.

RULE
Allowed-value condition. Field rule checks one value; record rule can compare fields.

VALIDATION TEXT
Helpful explanation shown when the rule fails.

MEMORY LINE
Mask = format; rule = condition; text = message.

Source Scope and Learning Links

The supplied Module 5 image lists these learning materials. The notes below cover the listed topics using the linked tutorials and Microsoft Support documentation. They are concise exam notes, not a verbatim transcript.

- Teacher's Tech, MS Access Tutorial for Beginners.
- Simon Sez IT, MS Access Forms Tutorial.
- Smyth Academy, Input Masks, Validation Rules and Validation Text.
- Microsoft Support, Control data entry formats with input masks.

- Microsoft Support, Restrict data input using validation rules.
- Learn Microsoft Access, Data Entry and Modification.